

Assistant	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness
Frontier API models						
Gemini 3.8 Flash	46	83	83	68	63	<u>79</u>
GPT-6 Astra	34	77	84	57	79	<u>73</u>
Gemini 3.1 Pro	31	71	78	61	68	69
Real-time streaming API models						
GPT-Realtime 2.1 (polled)	37	82	88	<u>67</u>	<u>77</u>	54
Gemini 3.8 Live (polled)	42	<u>90</u>	76	62	52	81
Open-weight non-streaming VLMs						
Muse Glimmer	26	70	<u>85</u>	63	76	45
Qwen3.8-27B	29	69	79	63	67	59
Kimi-VL-A3B-Thinking	12	56	60	42	37	31
InternVideo3-8B	18	64	62	45	24	44
Molmo2-8B	17	72	56	42	20	42
GLM-4.6V-Flash	<u>43</u>	91	68	55	46	77
Open-weight streaming VLMs						
Qwen3-Omni-30B	30	76	83	61	60	<u>79</u>
MiniCPM-o 4.5	16	62	59	35	39	29
Proact-VL	16	70	30	22	21	26
StreamingVLM	23	77	34	29	29	20

Table S Interaction quality in VHSim. Unaggregated scores for layout review. Scores are on a 0–100 scale (run 1). Overall: full 16-item CookSim rubric with mandatory-failure gating, not the mean of the five displayed dimensions. Dimension scores are averaged within episodes and then across episodes. Bold / underline: highest / second-highest per column. Row order follows CookSim success within each family.